



Workshop on the Use of Alternative Data Sources for Consumer Price Statistics

*Quality assurance,
validation, and comparison
with traditional price data*

 **UN Big Data Learning**

 **UN Big Data Regional Hub
in Brazil**

UN-CEBD Task Team
on Scanner Data

Learning Objectives

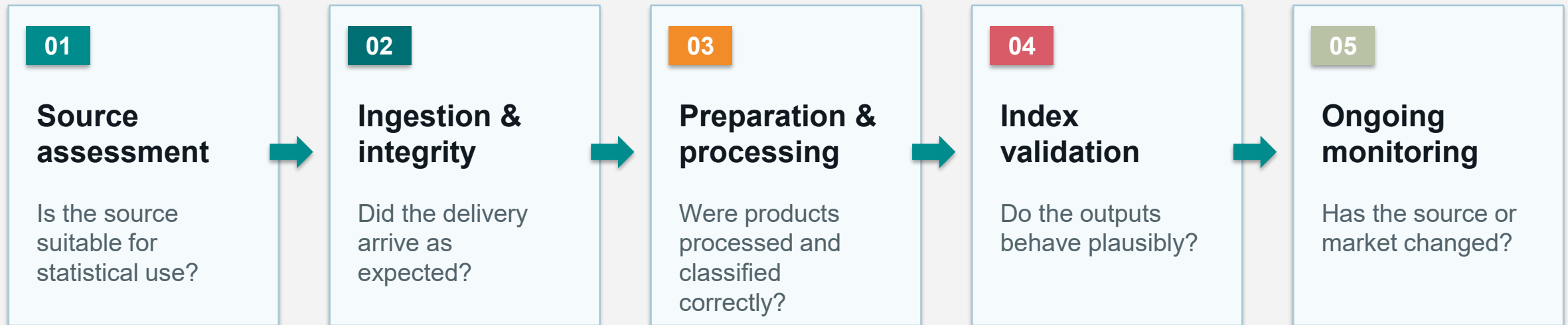
By the end of the session, participants will:

- . Be aware how quality frameworks can help structure this work
- . Understand the different quality assurance activities that can be considered across the CPI production process



QA across the CPI lifecycle

Quality is not a single validation stage. It is a set of control points that starts before data access and continues after publication.



Design principle

Match the depth of assurance to the scale, complexity and operational risk of the data source.



Quality frameworks

Quality Assurance
Framework of the
European Statistical
System

Relevance

Coverage and content meet CPI user needs.

Accuracy & reliability

Estimates are close to the target concept and stable across revisions.

Timeliness & punctuality

Data and outputs arrive when planned.

Accessibility & clarity

Metadata, methods and advice are sufficient for users.

Comparability & coherence

Series are comparable over time and coherent with related sources.

For modern systems (including use of AI) consider also reproducibility, explainability and robustness



Stage 1 — Assess the source

Decision to make: can this data source support CPI compilation reliably, sustainably and transparently?

Coverage fit

- 1 Products, outlets and market segments represented
- 2 Alignment with CPI concepts and definitions

Operational stability

- 1 Stable product identifiers and classifications
- 2 Reliable metadata and schema
- 3 Sustainable access to the source

Evidence base

- 1 Exploratory analysis or pilot study
- 2 Known limitations and risk profile
- 3 Decision record for approval or rejection

Practical output: a source assessment note that documents coverage, data definitions, metadata, access risks and conditions for use



Set data-sharing controls

Because alternative sources often sit outside NSO control, agreements should be agreed upfront:

Schema

Field names, definitions, types and allowed values.

Transfer frequency

Expected delivery timing and reference periods.

Delivery method

Channels, formats, encryption and failure procedures.

Responsibilities

Who maintains the supply and who resolves issues.

Change notice

Advance warning for system, definition or delivery changes.

Supplier validation

Checks performed before data reaches the NSO.

Ask upfront

“What changes could break our interpretation of the data, and how early would we know?”



Stage 2 — Integrity checks

The first production question is simple: did we receive the delivery we expected, in the structure we expected?

1	Receipt	Completeness of files or records received
2	Structure	Required variables present and correctly formatted
3	Identifiers	Key fields usable: product, outlet, date and transaction IDs
4	Volume	Observation counts within normal ranges
5	Anomaly flag	Sudden shifts, missing variables or schema changes investigated

Good practice: automate these checks quickly through a monitoring dashboard or report



Stage 3 — Assure processing

Large alternative datasets make automated processing essential; QA must therefore cover both data edits and the systems applying them

Map

Map of products to CPI classification

Clean

Missing and duplicate observations handled

Screen

Extreme or implausible price movements flagged

Maintain

Identifier changes, relaunches and pack-size changes managed

System controls

Reproducibility

Version control for code, data rules and method parameters.

Testing

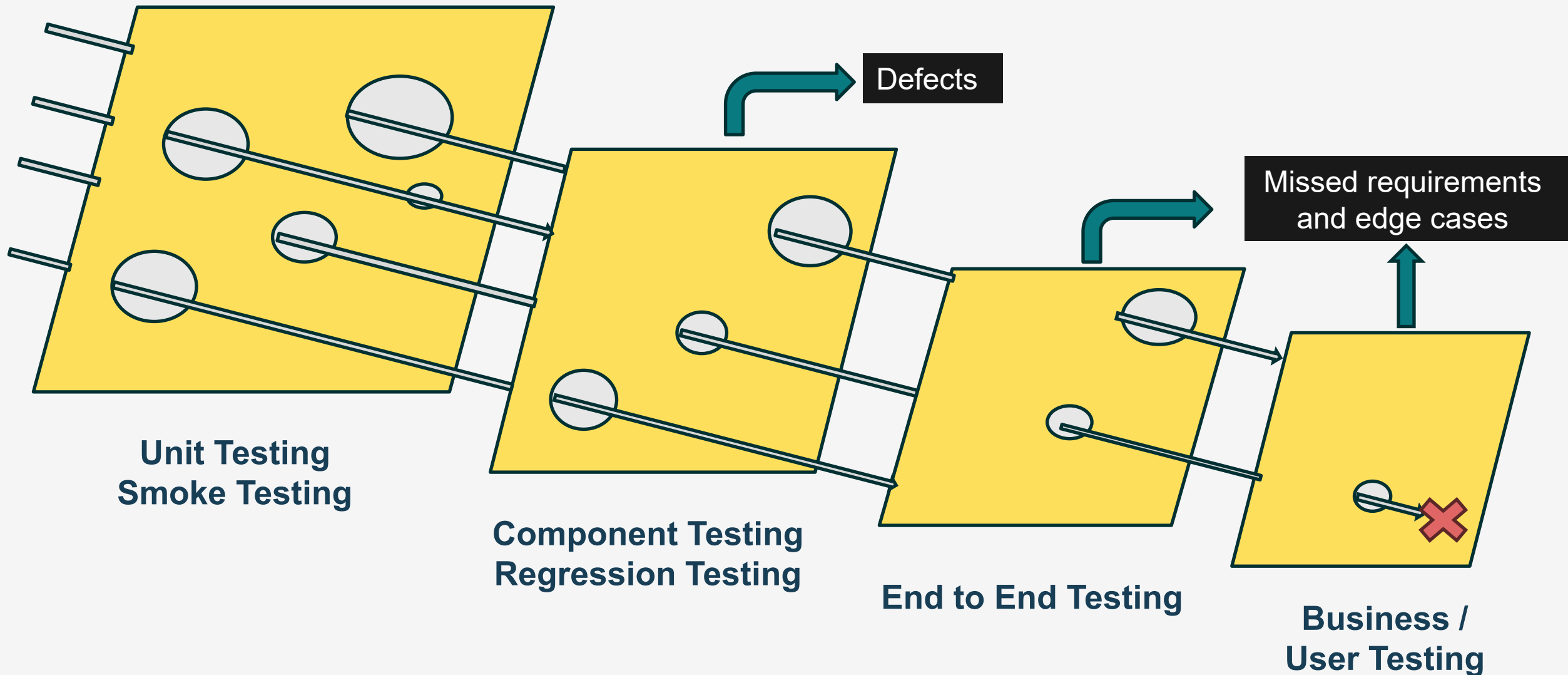
Unit, integration and regression tests for automated pipelines.

Ownership

Clear division of roles between statistical teams and IT support.



Types of Testing



Stage 4 — Validate indices

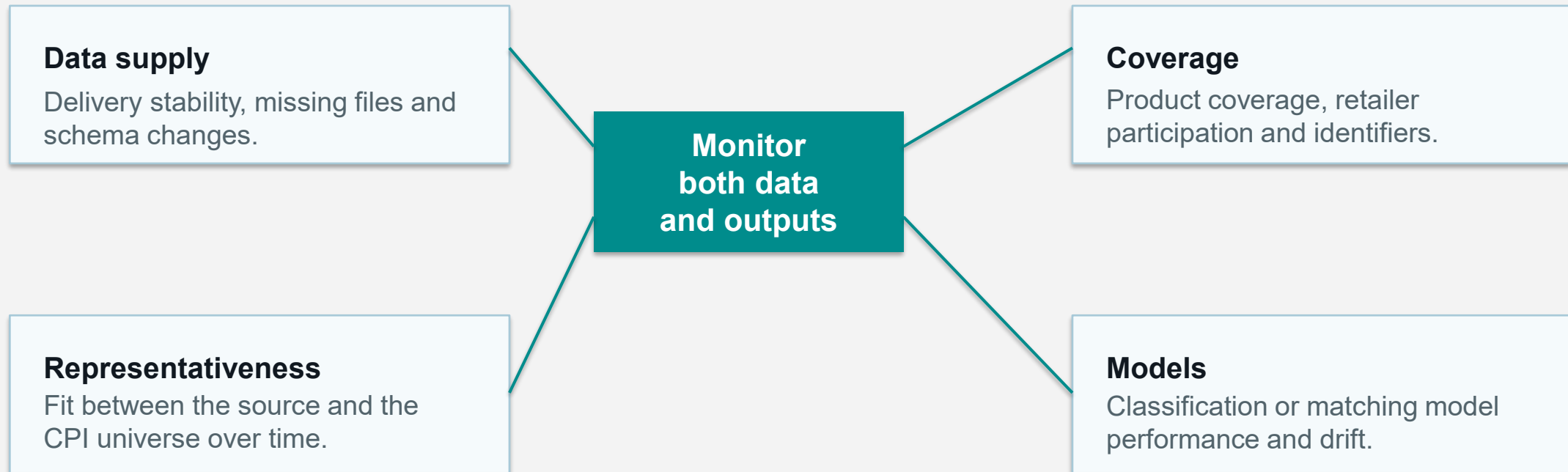
After compilation, QA shifts from record-level checks to the behaviour and explainability of statistical outputs

Signal in the index	Likely question	QA response
High volatility	Is this market movement or data noise?	Review products and retailers contributing most.
Structural break	Did coverage, filtering or matching change?	Trace method and input changes; document cause.
Different from CPI series	Is the new approach measuring a different basket?	Run impact analysis and explain differences.
Unexpected contribution	Is one retailer or product dominating?	Inspect weights, coverage and outlier rules.
Before live production: compare against existing CPI series, understand differences and document methodological changes for users.		



Stage 5 — Monitor over time

Alternative data sources evolve: retailers update systems, websites change and consumer purchasing behaviour shifts



Trigger for review: any sustained change in source characteristics, output volatility or model performance.



Minimum QA production pack

Use the workflow to define a repeatable production pack

1

Source register

Coverage, schema, provider dependencies and agreements.

2

Ingestion dashboard

Delivery status, structure checks and volume anomalies.

3

Transformation tests

Rules for classification, matching, filtering and outlier handling.

4

Validation pack

Volatility, contributions, comparisons and impact analysis.

5

Change log

Provider changes, pipeline changes and methodological decisions.

6

Governance route

Who reviews, approves and communicates changes.



QA is an operating model

For alternative CPI data sources, quality assurance works best when it is continuous, proportionate and documented

Embed

Place controls at every stage of the CPI lifecycle.

Automate

Use pipelines and dashboards to detect delivery and processing issues quickly.

Explain

Document differences, decisions and changes so outputs are transparent to users.



Practical example

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EXAMPLE WITH DOMINICK'S DATA



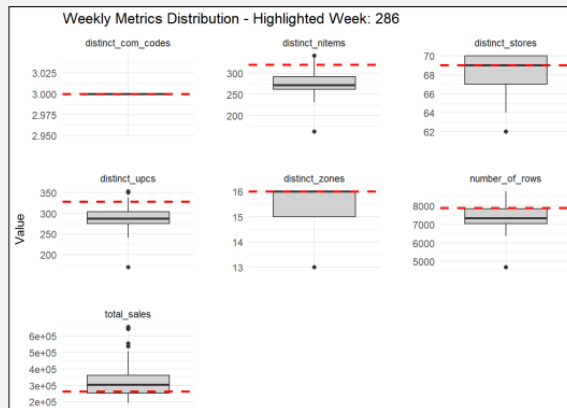
MONITORING / VALIDATION

See interactive examples online

Example weekly summary

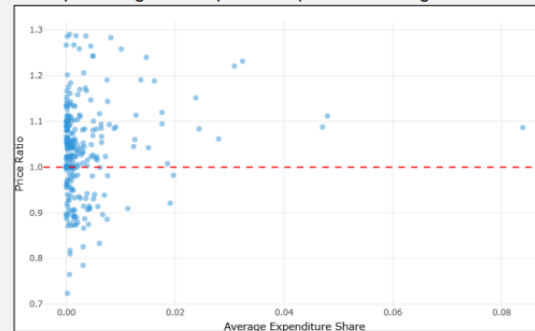
Column visibility	WEEK	number_of_rows	total_sales	week_start	week_end
	1	286	7880	263621.3	1995-03-02 1995-03-08
	2	287	8181	310166.82	1995-03-09 1995-03-15
	3	288	8098	296978.96	1995-03-16 1995-03-22
	4	289	8144	275520.61	1995-03-23 1995-03-29

Example statistics per week

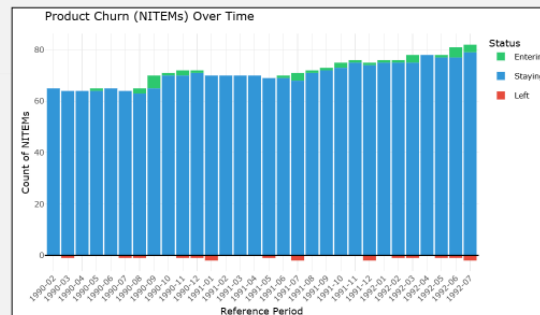


INPUT DIAGNOSTICS

Example change in unit price compared to average share

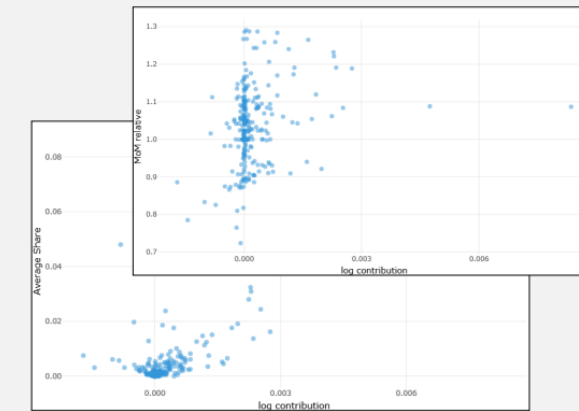


Example churn (entry, stay, exit)



ANALYTICAL / OUTPUT DIAGNOSTICS

Example overview of product contributions compared to their share in the index or the observed price change



Goussev, Serge, and Frances Krsinich. 2026. "Putting Windows in the Black Box: Developing a Set of Diagnostic and Analytical Tools to Support Multilateral Price Indexes in Production." *Presented at the 19th Ottawa Group Meeting, Warsaw, Poland*, accepted, June 10. <https://sergegoussev.github.io/multilateral-diagnostics/docs/paper.html>.

